

Filament Material Guide

High-Performance Filaments with HDT (2026 Reference)

Note: The values may be different depending on the brands - Check

Filament Name	Build Sheet	Cooling Fan Speed	Tg	Nozzle Temp	Bed Temp	Chamber Temp	Drying Temp	Drying Time	HDT	HDT-A	Bending Modulus MPa
ABS	PEI/Glass	Low	105	225	110	60	75	4	95	100	2300
ASA	PEI/Glass	Low	105	245	110	60	80	3	95	100	2000
CPE	PEI	Medium	90	250	80	20	60	3	70	75	1900
HIPS	PEI	0	100	225	100	50	-	-	80	80	1600
PA 6	G10/FR4	Low	105	265	110	70	90	4	140	180	2050
PA ePA CF	G10/FR4	Low	60	260	90	60	80	10	110	180	6500
PA12 CF	G10/FR4	Low	60	260	80	55	80	8	135	180	3000
PA12 GF	G10/FR4	Low	60	260	90	50	80	6	135	180	2900
PA6 GF	G10/FR4	Low	70	260	90	60	80	10	140	180	4800
PC	PEI	0	147	285	115	60	120	4	135	135	2200
PC + ABS		Low	137	275	110	80	110	4	126	128	2370
PC + ASA		Low	126	265	110	90	110	4	126	128	2370
PC BambuLab	PEI	Low	145	260	90	60	80	8	112	114	2300
PC Blend CF Prusa	PEI	Low	113	285	110	60	80	5	114	130	4200
PC BLEND LCF*		Low	158	280	100	90	80	6	185	185	9800
PC CF/GF		Low	143	280	110	100	80	5	135	140	6000
PC PBT CF		Low	140	260	100	100	80	5	110	110	6200
PC Polymax	PEI	Low	113	250	90	70	75	12	114	116	2040
PEEK	CF/PEI	Low	143	390	135	100	120	4	140	280	3700
PEKK	CF/PEI	Low	165	370	130	150	120	4	182	300	2900
PEKK CF	CF/PEI	Low	165	380	120	140	120	6	180	300	9850
PET CF	PEI	Low	70	260	70	55	80	10	112	225	6500
PETG	PEI	Medium	80	245	70	35	65	3	70	70	1600
PLA	PEI	High	85	210	23	25	65	3	60	100	3500
PPA CF	PEI	Low	90	300	90	60	100	8	227	250	9800
PPS	PEI	0	283	340	120	70	110	4	90	250	2540
PPS CF	PEI	0	100	320	110	80	100	10	264	280	7160
PPSU	PC/PEI	0	230	400	140	100	120	4	190	210	2215
PSU	FR4	0	187	365	130	130	120	4	172	180	2050
PVDF		Low	170	265	105	100	-	-	158	160	2100
TPC FLEX		Medium	80	235	70	35	65	4	120	120	100
ULTEM 1010	PC/PEI	0	186	375	135	110	120	4	208	210	3000
ULTEM 9085		0	180	360	130	170	110	5	177	180	2200

*Add North

NOTE:

Abrasive Protection:

Glass Fiber (GF) and Carbon Fiber (CF) will destroy brass nozzles.

A hardened steel or tungsten carbide nozzle is mandatory

Thermal Treatment:

Semi-crystalline polymers like PPS-CF and PPA-CF often require annealing to reach their maximum rated HDT

Active Drying:

These materials must be dried at 80°C–120°C for 8+ hours before printing to prevent structural defects

Amorphous Materials (PC, ABS, ASA, PETG, ULTEM):

These materials generally do not benefit from annealing in terms of HDT.

Annealing is used here primarily to reduce internal stresses and prevent warping later on.

Semi-Crystalline Materials (PLA, PPS, Nylon, PEEK):

Annealing is required to reach the high HDT values listed.

Without annealing, these parts will soften significantly at much lower temperatures.

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